# ACNE PHARMACOTHERAPY: Comparison Chart

Canadian'15'25, American-AAD'16'24, NICE'21'24, BMJ'25

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Acne affects 85% of those 12-24yr; duration varies ~4+ yr. Concerns include: scarring, pain, self esteem, social life, suicide. Contributing factors: hormonal, mechanical, contact, environmental, emotions, drugs. Family hx predictive of acne severity/duration.
Differential Diagnosis: rosacea, folliculitis, peri-oral dermatitis, keratosis pilaris, infections (bacterial, viral, fungal), tumors, milia. Aggressive tx early often preferred to prevent scarring in those at high risk.

**BASIC CARE**

- Avoid excessive skin care. Avoid stress, astringents, scrubs, toners.
- Use oil-free makeup. Don't pick/touch/squeeze lesions.
- **Shaving:** shave area lightly, only once & follow grain of hair growth.
- **Wash face:** preferably **once daily** & no more than BID with water & soapless cleanser (e.g., Cetaphil, Spectro Jel, CeraVe).
- **Strict sun protection**. Choose **non-comedogenic sunscreen** (e.g. Ombrelle Face, La Roche Posay Anthelios, Supergoop Matte) & **non-comedogenic moisturizers** (e.g. CeraVe). Tinted sunscreens (with iron oxides) are more effective than non-tinted sunscreens in preventing hyperpigmentation in patients with skin of colour.
- **Diet:** high fat / high sugar / high dairy may worsen; [42] limited evidence.

**Suggested Step-wise Approach for Initial Therapy**[3-10] **{Step-down in treatment intensity for maintenance following remission}**

Severity of Acne Mild → Severe

- **ISOtretinoin** (e.g. **ACCUTANE**) {Avoid with topical retinoid/BPO as ↑ drying effect}. **Intralesional corticosteroid injection** for larger acne papules/nodules.
- Systemic ABX (doxycycline) ± Topicals◊ {3 months of systemic antibiotics (up to 6 months[NICE'24]), then reassess/step down}
- ♀**Women**: CHC (e.g. **YAZ, YASMIN, DIANE-35**) ± Topicals◊. Spironolactone an option (stop meds if **pregnant**; 2 forms of birth control for ISOtretinoin)
- **If treatment failure:** escalate to combination topical therapy (i.e. **BPO ± retinoid ± antibiotic** → all three if necessary). May use combo tx (e.g. **CLINDOXYL, BIACNA, TACTUPUMP**), **clascoterone**, **azelaic acid**, &/or **salicylic acid**.† After treatment success, step down if possible.
- 1) **General measures** (see BASIC CARE); 2) Initiation of topical **benzoyl peroxide** (BPO), topical **retinoid** &/or topical **antibiotic** (topical antibiotic is not recommended as monotherapy). If affordable, may start combination therapy (but watch for increased skin irritation). Patient education important!

| Severity: | MILD (Comedonal) | Mild-MODERATE (Papulopustular) | SEVERE (Inflammatory) |
|---|---|---|---|
| Description[11] | <20 comedones (whitehead/blackhead), or <15 inflammatory papules, or lesion count <30. | 15-50 papules & pustules with comedones; cysts are rare; total lesion count may range from 30-125. | Primarily nodules & cysts; also present are comedones, papule & pustules, or total lesion count of >125. **Scarring**. |

**GENERAL APPROACH** for topical therapies: **Oily** skin → Use solution or gel; **Dry** skin → Use cream or lotions. **Potency of a given drug in various vehicles**: solution > gel > cream / lotion. Apply to **entire affected zone**, not just lesions!

## Acne – TOPICALS [12-14]

◊ **Oral** therapy plus adjunctive **topical** therapy [BPO ± Retinoids ± ABX] is beneficial. †**Practical Tip:** Use BPO/ABX at night (avoid BPO staining of clothing during day) with adapalene in AM (minimal sun concern).

| Generic/TRADE | Adverse Events AE / Contraindications CI / Drug Interactions DI / Monitor M / Comments | USUAL DOSE | $/pkg |
|---|---|---|---|
| **BENZOYL PEROXIDE (BPO)**: antibacterial/keratolytic. ✓ **1st line** for mild-moderate acne vulgaris as monotherapy. ✓Combo with other agents for mod-severe acne.† **Response time**: 8-12 wk. Clinical worsening may occur in first 2-4 wk. | | | |
| **Benzoyl Peroxide** = BPO, **g**. **OTC** strength ≤ 5%.<br>$H_2O$-based: **PANOXYL** 4% cream<br>**CLEAN & CLEAR PERSA-GEL** 5% gel<br>**OXY VANISHING** 5% lotion<br>**PROACTIV SOLUTION** 2.5% (system; $$$)<br>**NEUTROGENA STUBBORN ACNE** 2.5, 5% gel<br>Cleanser-based: **BENZAC W** 10% wash on SPDP,⊗<br>**BENZAGEL** 5% wash<br>**CeraVe** 4% Foaming Gel wash<br>Alcohol-based: **BENZAGEL** 5% gel ✗ ▼ | • **AE**: **Common**: irritation 50%, dryness 8%, & peeling 20% appear after a few days; erythema 14%; burning 1%; & pruritus 2%; may **bleach** hair/clothes; odour on clothing & bed sheets. **TITRATION**: ↓ irritation by: ↓ **concentration** (e.g. 2.5% vs 5%); ↓ **potency** ($H_2O$ < alcohol = acetone-based); or ↓ **frequency** (e.g. q2-3 days initially, or apply for 2 hr for 4 nights, 4 hr for 4 nights, & then leave on all night if tolerated). **Serious AE** (rare, 1-2%): Allergic reactions, contact dermatitis. Uncertain: BPO exposed to heat degrade to benzene=carcinogenic?.<br>• **DI**: **↑ skin irritation or drying effect** – concomitant topical medication, medicated abrasive soaps & cleansers, soaps & cosmetics with strong drying effect; products with high concentrations of alcohol, astringents, spices or lime; tretinoin (BPO oxidizes TRE, ↓ efficacy), isotretinoin.<br>• **BPO has oxidizing action** and may inactivate retinoids/antibiotics. Space admin times or use premixed combination products (examples below). BPO oxidizing action is antibacterial; combine with antibiotic (topical or oral) ↓ bacterial resistance (e.g. **BENZAMYCIN, BENZACLIN, CLINDOXYL**).<br>• **Efficacy**: BPO 2.5% has similar efficacy to 5-10%[limited evidence]; BPO 10% has more irritation (some drug plans). Avoid storing at high temperature.<br>• **Potency**: $H_2O$ < alcohol = acetone-based. Alcohol-based best if oily skin. Avoid washes & soap formulations → little residual contact time, less effect | Apply to entire affected area qHS or BID<br>(See titration comments at left.)<br>{**EPSOLAY**✗ ⊗ 5% cream microencapsulated $135/30gm}<br>**OTC**: 2.5, 4, & 5%<br>**Rx**: 8, 10, 15, 20% | **OTC**: $7-21<br>**Rx**: $15-25 |
| **RETINOID**: ✓ **1st line** for mild-moderate **comedonal** (blackheads/whiteheads) acne. ↓ noninflammatory & inflammatory lesions counts by 38-71%.[15] **Response time**: maximal at ~12 weeks. Clinical worsening may occur in first 2-4 wk. | | | |
| **Adapalene=ADA; Tretinoin=TRE; Tazarotene=TAZ; Trifarotene=TRI**<br>**ADA** PL LOW-POTENT; **TRE, g** PL MID-POTENT; **TAZ** PL HIGH-POTENT; **TRI** PL ?POTENT<br>↑ **potency** = ↑ **irritation**. Reserve **TAZ** for tough skin / need for strong tx<br>0.005%: **DIFFERIN**; **RETIN-A**; **TAZORAC**; **AKLIEF** $254 crm ✗ ⊗<br>0.01%<br>0.025%: $26 gel; $80 0.045% lot'n **ARAZLO**<br>0.04%: $107 **MICRO** gel ✗ ⊗<br>0.05%: $26 crm; D/C crm<br>0.1%: $159 crm,gel FDA:OTC; $107 **MICRO** gel ✗ ⊗; D/C gel<br>0.3%: $243 **XP** gel ✗ ▼ | • **AE**: **Common**: **skin irritation** i.e. red, dry, burning, hypopigmentation, **photosensitivity**. **Rare**: contact allergy. **CI**: eczema at site; sunburn. **Caution**: pregnancy; ↑ risk if >20% BSA, skin lesion, etc.<br>• No antibacterial effect; combination with ABX does not prevent resistance.<br>• **Pediatrics**: limited data for age <12 yr.<br>• Retinoids ↓wrinkles & post-acne scars.[41]<br>• **DI**: TRE+BPO (↓efficacy) except combination **TWYNEO**<br>• **Efficacy**: ADA similar efficacy to BPO in ~12 wk.[Cochrane'20] | **Table 1. Management/Prevention of Retinoid Skin Irritation**<br>• Ensure skin is completely dry before applying (↑irritation if damp skin).<br>• Start at a low strength &/or every other night; gradually titrate. If needed, ↓ contact time (e.g. 5-15min, then wash off). Creams ↓ irritation vs gels.<br>• **To ↓photosensitivity**, apply at HS & ensure **sunscreen use**. Consider stopping retinoid 1 week before a sunny vacation.<br>• **Moisturize** twice daily; glycerin-containing moisturizer ↑ hydration.<br>• Adapalene and **RETIN-A MICRO** generally best tolerability (but ↑cost).[44]<br>• After response to therapy, consider ↓ frequency (e.g. q2-3 nights).<br>• Tolerability generally improves over time (e.g. after 1-3 months).<br>• Avoid meds/products that ↑drying (e.g. astringents, abrasive soaps, isotretinoin). | See **Potency Table** for prices.<br>Apply pea-sized amount **qHS** (should disappear within 1 min). Wait 30-45 min after wash; start low %; apply **q2-3 nights** initially to ↓AE.<br>May give **ADA** in **AM** due to less photosensitivity. | |
| **ANTIBIOTIC**: ✓ Tx of mod-severe acne. Most effective for inflammatory lesions (**stop** if/when no further inflammation). Prevent resistance by avoiding **monotherapy**; consider combos below instead. **Response time**: 8-12 weeks. | | | |
| **Clindamycin** = CLI<br>**DALACIN T**, **g** 1% topical solution<br>**CLINDETS** 1% topical pads<br>**Erythromycin** = ERY **ERYSOL** 2% gel + SPF-15 D/C. | • **AE**: **Common**: erythema, peeling, itching, dryness & burning (but less irritating than BPO & TRE).[16] **Serious** (rare): pseudomembranous colitis. **CI**: clindamycin: previous colitis, ulcerative colitis, pseudomembranous colitis, regional enteritis. {**CLINDASOL** 1% cream + SPF-15 **D/C**.}<br>• Most effective when used in combination with BPO or topical retinoids.[17-20] **Combination with BPO prevents bacterial resistance**.[21-23]<br>• Clindamycin may be preferred over erythromycin for prolonged effect &/or less resistance.[Expert Opinion] | CLI 1% soln, **g**: BID<br>**CLINDETS**: BID<br>ERY: daily-BID | $38 **g** 60mL |
| **Azelaic Acid** **FINACEA** 15% gel, 20%crm AZELEX | ↓ irritation vs BPO but ↓ effect;[45] OK for mild acne; onset ~4wk; hypopigmentation can occur, esp. darker-skin. Combo OK with BPO or retinoid. | Apply daily to BID | $55 50g |
| **Dapsone** **ACZONE** 5% gel ✗ ⊗, 7.5% gel | P1,2 P2 L Effective ≥12yr esp inflammatory acne in ♀; **AE**: discolor skin yellow/orange with BPO; ↑ skin AE if HLA-B*1301 positive . **CI**: G6PD (methemoglobinemia) | Apply BID | $102 60g |
| **COMBINATIONS**: ✓ Tx of mod-severe acne.† Combos ↑ efficacy vs monotx (e.g. BPO + CLI 38% clearance vs BPO alone 30%, **NNT≈12**).[47] After inflammation settles, try stopping PO antibiotic! **Response**: ~ 2-4weeks, optimal 8-10 weeks. | | | |
| **BENZAMYCIN** = BPO 5% + ERY 3% gel ❄<br>**BENZACLIN, CLINDOXYL, g** = BPO 5% + CLI 1% gel; **CLINDOXYL ADV** = BPO 3% + CLI 1% gel<br>(**Clindoxyl** ❄ prior to dispensing, then dispense at RT). **STIEVAMYCIN** = TRE + CLI: D/C.<br>**BIACNA** = TRE 0.025% + CLI 1% gel ✗ ▼<br>**TACTUPUMP, g** = ADA 0.1% + BPO 2.5% gel age ≥9yr {**Forte, g** ADA 0.3% + BPO 2.5% ⊗ $110 g,$219 age ≥12yr}<br>**CABTREO** = CLI 1.2% + ADA 0.15% + BPO 3.1% gel age ≥12yr | • **AE, CI, DI**: as for individual ingredients above.<br>• **TACTUPUMP** combo superior to either ingredient alone (**NNT≈10-17**), but ↑ irritation.[47]<br>• Keep **BENZAMYCIN** in refrigerator (3 month expiry); **CLINDOXYL** at room temp (2 month expiry).<br>• **BIACNA** for comedones & inflammatory lesions age ≥12yr; resistance concern if used long term.<br>• **CABTREO** once daily. Refrigerate in pharmacy. Pt: store at room temp, use within 10 wk after open. | **BENZAMYCIN**: qHS-BID<br>**BENZACLIN**: qHS-BID<br>**CLINDOXYL**: qHS-BID<br>**BIACNA**: qHS<br>**TACTUPUMP**: qHS<br>**CABTREO**: qHS | $85 46.6g<br>$43 g, $75 50g<br>$44 g, $48 45g<br>$106 60g<br>$70 g, $138 70g<br>$172 50g |
| **OTHER:** **Tea tree oil** 5%: 1 small efficacy trial but relatively slow onset.[24] **Sulfur & Resorcinol**: weak efficacy. **Chemical peels** trichloroacetic acid, glycolic acid, salicylic acid. **Home LED red/blue devices**: ↓ mild-moderate acne lesions. | | | |
| **Clascoterone** **WINLEVI** 1% cream ✗ ⊗ PL | Anti-androgen studied in ♀ & ♂; efficacy in 18-20% of pt, **NNT≈8-11**/12 weeks. Effective for inflammatory and noninflammatory acne.<br>**AE**:mild skin reactions - red/dry skin, scaling, pruritus, stinging, burning, edema, adrenal suppression 5%.[48] | Apply BID age ≥12yr | $300 30g |
| **Salicylic Acid** 0.5, 1, 2, 3.5% crm/sol'n ✗ ⊗ PL | **OXY, CLEARASIL, NEUTROGENA, other brands**. Less potent than equal strength BPO; option if retinoid failure. **AE**: allergic reactions, red skin. | Apply daily to BID | $10-15 |

**Additional Topicals**: BELLAFILL FDA dermal filler: useful to correct scarring. **AMZEEQ** minocycline 4% topical foam daily in ≥9yr; **AE**: H/A. **TWYNEO** tretinoin 0.1%/BPO 3% cream daily (HS), $470USD/30g (HC approved 2025, not yet marketed).

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| Generic/TRADE | Adverse Events AE / Contraindications CI / Drug Interactions DI / Monitor M / Comments | USUAL DOSE | $/90d |
|---|---|---|---|
| **ORAL ANTIBIOTICS**: Targets *Cutibacterium acnes* (aka *P. acnes*). ✓ mod-to-severe acne; ✓ acne on the chest, back or shoulders; ✓ pt with inflammatory disease in whom topical combinations have failed or are not tolerated; ✓ moderate acne with tendency for scarring or substantial post-inflammatory hyperpigmentation. More effective than CHC at 3 months, but not at 6 months. TET, DOX, & MIN = 50-60% rate of improvement in inflammatory lesions after 8 wk;[53] ERY = 67% **(2nd line**, see below).[25] TRI **3rd line** for when other ABX are CI. **"Pulse Therapy"**: po ABX for 3-4 months plus topical, then step down to topical therapy alone. Recommend BPO to help ↓ABX resistance (avoid topical abx monotherapy). **Short tx courses** (≤12 wk) help ↓resistance. If unable to discontinue oral ABX, consider using the lower maintenance dose. **Lack of response**: may relate to resistance, especially with ERY; less with TET, DOX, or MIN. | | | |
| **Tetracycline** = TET, g<br>TETRACYN 250mg cap | • **AE**: **Common**: GI upset 7% (TET > MIN or DOX),[43] vaginal candidiasis, photosensitivity (DOX > TET > MIN; esp. UVA, & dose-dependent i.e. less of a problem at DOX 100mg/day). Use Sunscreen SPF ≥ 30. MIN: hyperpigmentation (grey-blue, irreversible) of skin & mucous membranes, lightheadedness, dizziness, vertigo, ataxia, drowsiness & fatigue. **Serious**: rare azotemia, pseudotumor cerebri (benign intracranial hypertension). MIN: rare lupus-like reaction, autoimmune hepatitis & hypersensitivity syndrome (some suggest avoid), spermatogenesis may affect, ?thyroiditis.<br>• **CI**: Pregnancy (unless strongly indicated e.g. lyme dz), children <8yr, severe renal or hepatic dysfx; DOX: myasthenia gravis (possible association with muscle weakness)<br>• **DI**: ↓ GI absorption: $Fe^{++}$, bismuth, $Al^{+++}$, $Ca^{++}$, $Mg^{++}$ (separate dose by 2 hr); ↑INR: warfarin; may ↓ oral contraceptive effectiveness; isotretinoin (intracranial HTN/hemorrhage).<br>• **M**: MIN: consider LFTs & antinuclear factor baseline & q3-4 months.<br>• **Take TET on empty stomach** - absorption is ↓ by food & dairy. **No TET before sleep** as pills may lodge in the esophagus & cause ulceration.<br>• DOX & TET have cross-resistance (but not MIN). **DOX**: ➡ **advantage** of convenient daily dosing without the AEs, or monitoring of MIN.<br>• 🇺🇸 **SEYSARA sarecycline** 60-100-150mg tab oral daily mod-severe non-nodular acne ≥9yr; **AE**: hypersensitivity, tooth discolor; **DI**: digoxin. $**1015**/mo | *Initial*: 500mg BID **ac**<br>*Maint*: 250-500mg **ac** daily | $66<br>$40-50 |
| **Doxycycline** = DOX, g<br>DOXYCIN 100mg cap, tab | | 100mg daily-BID<br>**ac best**, but may take cc if GI upset | $77-124 |
| **Minocycline** = MIN, g<br>MINOCIN 50mg, 100mg cap | | *Initial*: 100mg daily<br>*Maint*: 50-200mg/d<br>May take cc | $137<br>$86-245 |
| ~~**Erythromycin** = ERY, g~~ **Discontinued**<br>ERYC, ERYBID 250mg, 333mg, 500mg, others | • **2nd line** due to ↑ resistance and GI effects. **AE**: **Common**: GI upset, N/V/D, vaginal candidiasis. **Serious** (rare): estolate-induced cholestatic jaundice. **CI**: ERY estolate - pre-existing liver dx. **DI**: Inhibits CYP1A2 & 3A4: ↑ levels of carbamazepine, cyclosporine, theophylline, warfarin. | *Initial*: 500mg BID<br>*Maint*: 250-500mg daily | $93<br>$31-51 |
| **Trimethoprim** = TRI, g<br>PROLOPRIM 100mg, 200mg tab | • **3rd line** only. **AE**: **Common**: GI upset, rash 3% usually self-limiting. **Serious** (rare): hepatic/renal toxicity, agranulocytosis, TEN.<br>• **CI**: folate deficiency (anti-folate properties can worsen megaloblastic anemia). In women of childbearing potential, give folic acid 1mg/d.[SOGC1A] | 100mg BID or 200mg daily (max: 600mg/d) | $96 |
| **COMBINED HORMONAL CONTRACEPTIVES (CHC)**: Anti-androgenic effect due to estrogen ↑SHBG. Progestin choice may or may not be important.[26] ✓ ♀ with mod-to-severe acne ± seborrhoea ± hirsutism ± androgenic alopecia ± late onset acne ± desiring contraception. Overall > 50% improvement in lesions, **NNT≈4-9/~24 wk** in mild-moderate acne. Relapses common after D/C.[36] **Response time**: 3-6 months. Clinical worsening may occur early in cycle. | | | |
| TRI-CYCLEN, g EE 35mcg + norgestimate 0.18/0.215/0.25mg<br>ALESSE, g EE 20mcg + levonorgestrel 0.1mg<br>DIANE-35, g EE 35mcg + cyproterone 2mg ✗▼<br>YASMIN, g EE 30mcg + drospirenone 3mg<br>YAZ, g EE 20mcg + drospirenone 3mg<br>YAZ-PLUS EE 20mcg + drospirenone 3mg + levomefolate 450mcg ✗▼ | • See RxFiles Oral Contraceptive Chart, page 176 for full list of **AE**, **CI**, **DI**, & **M**. Norethindrone CHC can exacerbate acne.<br>• **AE**: **Common**: breakthrough bleeding, HA. **Serious** (rare): cyproterone - hepatotoxicity, **VTE** (3.4/10,000 ♀-yr in 1st yr; ?risk may be ↑ with drospirenone), ↑$K^+$ (rare - drospirenone). **CI**: smoking, DVT risks, migraine with aura. **DI**: ?**Oral antibiotics** may ↓ CHC efficacy.<br>• **Efficacy**: All CHC beneficial (but only listed agents have official indication in Canada). YASMIN as effective as TRI-CYCLEN & DIANE-35.[27,28] Cyproterone or drospirenone anti-androgenic **may** have ↑ effect (but evidence limited).[26] Progestin-only contraceptives may worsen acne.<br>• DIANE-35 lacks contraception indication in Canada (but effective); has this indication in other countries (e.g. Australia). | TRI-CYCLEN<br>ALESSE<br>DIANE-35<br>YASMIN<br>YAZ<br>YAZ-PLUS<br>Once daily (cycle) | $63 g<br>$28 g<br>$97 g<br>$36 g<br>$42 g<br>$58 |
| **ORAL RETINOID**: **Most effective** acne treatment (↓ sebum, comedone formation, *C. acnes*, & inflammation)[29] but significant AE - **reserve use**. ✓ Severe nodulocystic acne; ✓ acne associated with scarring; ✓ failure to respond / intolerability of prior systemic therapy; ✓ significant psychological distress because of acne; ✓ acne fulminans, gram-negative folliculitis, or pyoderma faciale.[30] Remission rates as high as 70-89%.[31-33] **Response time**: ~3 months. May see initial flare. | | | |
| **ISOtretinoin** = ISO, g<br>ACCUTANE 10mg, 40mg caps soybean 1-888-762-4388<br>CLARUS 10mg, 40mg caps soybean 1-877-776-7711<br>EPURIS 10mg, 20mg, 30mg, 40mg caps soybean<br>ABSORICA LD ✗⊗ 8, 16, 24, 32mg cap micro soybean<br>♀: **Test for pregnancy twice before (once at initial assessment & the other within 11 days prior to initiating), during (monthly), & after discontinuation (1 month).** (↑ risk CNS, ears, eyes, heart defects)<br>◆ **Initiate** 2-3 days after next normal menstrual period<br>◆ **2 reliable contraception forms required,** unless abstinence is chosen method; continue until 1 month after discontinuation<br>◆ **Not a major issue for males/sperm**<br>**Web:** www.claruscleanprogram.ca | • **AE**: AE dose related; consider low dose & slow titration.[34,13] Note: low doses ↑ risk of relapse if **optimal cumulative dose** not achieved. **Common**: **dry skin & mucous membranes** (e.g. skin 80%, lips 93%, mouth 33%, eyes 35%, nose 80%, bleeds 20%, peeling of fingertips 20%, itching 41%, thirst 30%), hair loss, rash/red face 34%, **aches** (e.g. HA 13%, myalgia, back pain 5%; tx with acetaminophen or NSAID) ↑ lipids (e.g. from baseline: chol↑20%, LDL↑>15%, TG↑>5.7mmol/L in 25% of pts & pancreatitis risk↑, HDL↓15%), ↑ **sun sensitivity** (use sunscreen SPF ≥30, esp UVA) ↑ LFTs usually transient. Dryness worst in first 8 weeks → tx with **lip balm**, temporary removal of contact lens; **eye lubricants**, Vaseline or **nasal** moisturizers (e.g. Rhinaris/ Secaris), 1g/day oral omega-3.[50] **Avoid other acne topicals** during ISO therapy due to ↑dryness. **Serious**: abrupt ↓ night vision (stop ISO), depression & suicide controversial: no direct evidence & untreated acne ↑psych risk, but monitor,[35] ?IBD, ?SJS/TENS, ?sexual dysfx, ?sacroiliitis, DISH case reports.[53]<br>• **CI**: Pregnancy, hyperlipidemia, ↑Vit. A, soybean allergy (peanut allergy?), <12yr, eye-LASIK surgery. **DI**: CHC, methotrexate, tetracycline, Vit. A.<br>• **M**: CBC, LFTs, lipids: 0 and after 8 wk; **pregnancy tests** (see side bar), mood. Monitoring form: rxfiles.ca/rxfiles/uploads/documents/Isotretinoin-monitoring.pdf<br>• Lesions localized on the face, upper arms & legs tend to clear more rapidly than trunk.[36]<br>• **Initial acne flare up** may occur during the first 2 months of tx (~6% of patients).[37] If flare severe: stop ISO, restart at lower dose of 0.1mg/kg/d, & slowly ↑ to 0.5mg/kg/d; OR give prednisone 0.5-1mg/kg/d x 2-3 wk with a gradual taper.<br>• **Improvement persists 1-2mo after stopping, but relapse may occur** after therapy completion. To prevent relapse, total optimal cumulative dose = **120-150 mg/kg/course** (leads to 55-80% long-term remission). (Recent evidence suggests further ↓ in relapses with ≥**220mg**/kg/course.)<br>• **PK**: $t_{1/2}$ = 10-20hr; ↑ absorption with high fat meal (take **cc;** however **ABSORICA** is micronized and can be taken without food)<br>• **CLARUS** 40mg caps most cost-effective; however **EPURIS** 20 & 30mg caps offer dose flexibility and ↓ cost vs **CLARUS**/**ACCUTANE** 10mg caps.<br>**Alternative doses:** Micronized formulation: 0.4-0.8mg/kg/day in 2 divided doses, up to 1.6mg/kg/day for severe disease/scarring. Continue therapy until nodule count ↓ by at least 70% or up to 20 weeks. Low dose - moderate acne: 0.3-0.5mg/kg once daily, total course=120-150mg/kg. | *Initial*: 0.5mg/kg/day **cc** divided daily-BID x4wk<br>*Maint*: 1mg/kg/d **cc** x3-7 months.[38] Max: 2mg/kg/d.<br>Total course: 120-150mg/kg<br>**E.g. 60kg (40mg caps)**<br>• 40mg daily x 4wk;<br>• 40 & 80mg alternating daily x 4 months<br>• Total course = 130mg/kg<br>**E.g. 60kg (20, 30mg caps)**<br>• 20mg BID x 4wk;<br>• 30mg BID x 4 months<br>• Total course = 130mg/kg<br>**ACCUTANE** 10mg caps = $**870**/course | Suggest limit to 1 month supply.<br>**CLARUS** 40mg cap $460 / course<br>**EPURIS** 20, 30mg $**790** / course |
| **SPIRONOLACTONE**: Anti-androgenic. Off-label use: acne in adult women ♀ ≥18yr when other treatments have been ineffective, not tolerated, or CI. **Response time**: up to 1 yr (82% response vs 63% placebo; **NNT≈5-6/~24 weeks**).[49] | | | |
| **Spironolactone**, g<br>ALDACTONE 25mg, 100mg tab | • **AE**: **Common**: menstrual irregularity, mild GI upset, headache, ↑$K^+$ (check baseline levels), gynecomastia, breast tenderness.<br>• **CI**: anuria, CrCl <30mL/min, hyperkalemia. **M**: $K^+$ if age >45yr, ↓CrCl, or on other meds that impact $K^+$.[46,52] | 25-200mg daily (Usually 50-100mg) | $15-29 ($18-20) |

**Other Systemic Meds**: **Clindamycin (oral) DALACIN C** & cotrimoxazole **BACTRIM, SEPTRA** not commonly used → pseudomembranous colitis & TEN, respectively.[37]
**Azithromycin ZITHROMAX** 250mg 3x/wk is being used in acne, but studies are preliminary.[32]
**Prednisone PRED** 2.5-7.5mg or **dexamethasone DEX** 0.125-0.5mg qHS for congenital adrenal hyperplasia or temporary benefit in severe inflammatory acne.
**Flutamide** 125-375mg/d for hirsute females x 1-6 mo (but potential hepatic toxicity).
**Triamcinolone** 0.25-0.5mg injected into inflammatory cysts, larger acne papules or nodules for acute cosmetic purposes.

**Drug induced Acne**: **anabolic steroids**, androgens in women, contraceptives high in progestin, corticosteroids, corticotrophin (ACTH), bromides, cetuximab, chlorides, coal tar topical, crystal meth, cyanocobalamin, cyclosporine, dantrolene, dioxins chloracne, erlotinib, gabapentin, gefitinib, gold salts, halothane, iodides, isoniazid, JAKi, **lithium**, medroxyprogesterone, panitumumab, phenobarbital, phenytoin, psoralens, tacrolimus, quinidine, quinine.

**Pregnancy:** Topical azelaic acid, BP, clindamycin, erythromycin and salicylic acid are considered safe in pregnancy,[BMJ'25, CAN'25] **STOP** all other acne medications if pregnant.

# ACNE

Abbreviations

✗ =Non-formulary in Sk ☎=Exception Drug Status SK ⊗=not covered by NIHB ▼=covered by NIHB ✆=prior approval NIHB **ABX**=antibiotic **Al$^{+++}$**=aluminum **Ca$^{++}$**=calcium **chol**=cholesterol **crm**=cream **D/C**=discontinued product **DISH**=diffuse idiopathic skeletal hyperostosis **EE**=ethinyl estradiol **esp.**=especially **Fe$^{++}$**=iron **$H_2O$**=water **IBD**=Inflammatory bowel disease **JAKi**=janus kinase inhibitor(s) **K$^{+}$**=potassium **LED**=light-emitting diode **Mg$^{++}$**=magnesium **mo**=month **N/V/D**=nausea/vomiting/diarrhea **OTC**=over-the-counter **PK**=pharmacokinetics **Rx**=prescription **SHBG**=Sex hormone-binding globulin **TCN**=tetracycline(s) **TEN**=toxic epidermal necrolysis **TG**=triglyceride **tx**=treatment **Vit. A**=Vitamin A **wt**=weight

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## Other acne drugs:

| | | |
|---|---|---|
| **Salicylic Acid** = SA▼✗<br>**Oxy, Clearasil, Neutrogena, others**<br>Gels, lotions, toners, cleansers, sticks, pads, washes & astringents<br>0.5, 1, 2 & 3.5% PL | **Common:** less irritating than BPO, burning, stinging, pruritus & erythema<br>**Serious**: rare systemic salicylate toxicity: nausea, vomiting, diarrhea, dizziness, loss of hearing, lethargy, psychic disturbances & hyperpnea<br>?protect from sun<br>8-12 weeks for noted improvement | ✓ Used with topical retinoids to treat mild comedonal acne or 2nd line monotherapy agent[3] (also for seborrhea & psoriasis).<br>**Not commonly recommended** (less potent than equal strength BPO).<br>**D**: ↑ **skin irritation or drying effect**: Abrasive or medicated soaps or cleansers; Acne preps (e.g., BPO, Resorcinol, Sulfur, Tretinoin); alcohol-containing topicals (After-shave lotions, perfumed toiletries, cosmetics/soaps with a strong drying effect); Isotretinoin DAILY or BID, 3-6% is keratolytic , **OTC**: $**10-15** |

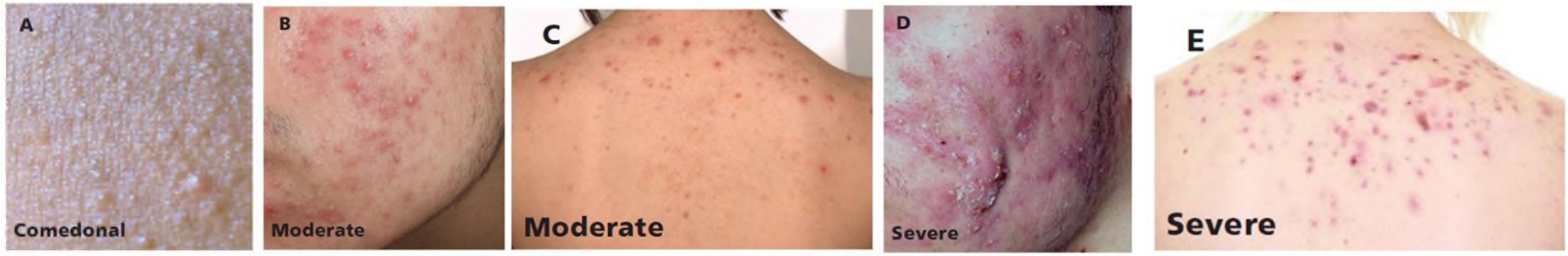
**Methylprednisolone/Neomycin/colloidal Sulfur** NEO-MEDROL ACNE $^{p/c}$ 0.25%/0.25%/5% lotion; daily-BID $24/75mL ✗ ⊗; may exacerbate acne; **AE** (corticosteroid): burning sensation, itching, irritation, dryness, folliculitis, acneiform eruptions, hypopigmentation; rare true contact allergy. – not available

**Sulfacetamide sodium/colloidal Sulfur** SULFACET-R 10%/5% lotion; BID-TID; less efficacious for acne, but useful for acne rosacea; tinted prep may be useful as camouflage. – not available

**INVESTIGATIONAL**: **Bimekizumab** (off-label) for acne conglobate (case reports)

**Tetracycline Lactation Ratings:** a more conservative approach was used for acne (i.e. safe/likely safe changed to caution) as lactation data is only available for short-term courses versus the 8-12 weeks of therapy for acne.

## Determining severity of acne (examples):




Reference: Canadian 2015 Acne Guidelines (Asai et al)

### Pregnant Women With Acne[BMJ'25]

- Topical clindamycin, erythromycin and azelaic acid are considered safe in pregnancy women with acne.[BMJ'25]
- Some evidence suggests salicylic acid can be used be used in pregnancy if the area of exposure and duration of therapy is limited. It is not recommended for use in large areas or under occlusion due to the potential for systemic absorption.
- NOT recommended in pregnancy: Oral macrolides, spironolactone, oral contraceptives, ISOtretinoin.

**Glossary of lesion terminology**

| Lesion | Definition | Size (if applicable) |
|---|---|---|
| **Comedo** | Blocked follicle. | |
| **Open comedo ("blackhead")** | Follicle in which keratin and sebum accumulates and often presents as a flat or slightly raised lesion with a darker coloration with a central pore. | |
| **Close comedo ("whitehead")** | Whitish papule without a central pore and without inflammatory signs. | |
| **Papule** | Elevated lesions | < 1 cm |
| **Pustule** | Elevated lesions | < 1 cm with purulent content. |
| **Nodule** | Elevated lesions | > 1 cm |
| **Cyst** | Elevated lesions | > 1 cm, fluid filled |

**Acne vulgaris vs. drug-induced acneiform eruptions**

| Feature | Acne Vulgari | Drug-induced acne |
|---|---|---|
| **Lesion characteristics** | Polymorphous lesions: comedones, papules, pustules, nodules and cysts; multiple stages of evolution of lesions simultaneously. | Monomorphous lesions: inflammatory dome-shaped papules and postules resembling rose pearls (pinkish erythematous lesions); absence of comedones is typical. |
| **Symptoms** | Not pruritic | Typically pruritic |
| **Location** | Sebhorrheic regions: predominantly on the face and to a lesser extent on th trunk, chest, neck, and proximal areas of upper extremities. | Widespread: back and chest, possibly affecting the face and upper extremities; may also appear on lower back and genitalia, but not predominantly in seborrheic areas. |
| **Age and onset** | Typically during adolescence, but may appear in adulthood. | All ages, often without a history of acne. Time of onset depends on the offending drug.<br>JAKi: sudden onset, associated with treatment initiation.<br>Steroid: latency period of 2-4 weeks. |

**Search Terms**

| | |
|---|---|
| Acne | 33 |
| ACZONE | 33 |
| Adapalene | 33 |
| AKLIEF | 33 |
| AMZEEQ | 33 |
| ARAZLO | 33 |
| Azelaic acid | 33 |
| BENZAC W | 33 |
| BENZACLIN | 33 |
| BENZAGEL | 33 |
| BENZAMYCIN | 33 |
| Benzoyl Peroxide | 33 |
| BIACNA | 33 |
| BPO | 33 |
| Cetaphil | 33 |
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| PROACTIVE | 33 |
| RETIN-A | 33 |
| Soap | 33 |
| Spectro Gel | 33 |
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| | |
|---|---|
| STIEVAMYCIN | 33 |
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| Tazarotene | 33 |
| TAZORAC | 33 |
| Tea Tree Oil | 33 |
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| ACCUTANE | 34 |
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| ALDACTONE | 34 |
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| TWYNEO | 33 |
| EPSOLAY | 33 |

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**Other Articles of Interest**:

**2026**:

Schneeweiss MC, Wyss R, Anand P, et al. Risk of Acne in **Patients With Atopic Dermatitis** Starting a JAK Inhibitor vs Th2 Cytokine Inhibitor. JAMA Dermatol. 2026 Jan 21:e255443.
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Schmidt SAJ, Grønborg TK, Gribsholt SB, et al. **Isotretinoin Treatment** in Adolescence and Adult Height. JAMA Dermatol. 2026 May 20:e261197.

**Prior to 2026:**

**AADA** Nov/10 American Academy of Dermatology Association (AADA) updated its position statement on the use of **isotretinoin**. http://www.aad.org/forms/policies/Uploads/PS/PS-Isotretinoin.pdf
Abdel Hay R, Shalaby K, Zaher H, et al. **Interventions for acne scars**. Cochrane Database of Systematic Reviews 2016, Issue 4. Art. No.: CD011946. DOI: 10.1002/14651858.CD011946.pub2. There is a lack of high-quality evidence about the effects of different interventions for treating acne scars because of poor methodology, underpowered studies, lack of standardised improvement assessments, and different baseline variables. There is moderate-quality evidence that injectable filler might be effective for treating atrophic acne scars; however, no studies have assessed long-term effects, the longest follow-up being 48 weeks in one study only. Other studies included active comparators, but in the absence of studies that establish efficacy compared to placebo or sham interventions, it is possible that finding no evidence of difference between two active treatments could mean that neither approach works. The results of this review do not provide support for the first-line use of any intervention in the treatment of acne scars. Although our aim was to identify important gaps for further primary research, it might be that placebo and or sham trials are needed to establish whether any of the active treatments produce meaningful patient benefits over the long term.
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Arowojolu AO, Gallo MF, Lopez LM, Grimes DA. **Combined oral contraceptive** pills for treatment of acne. Cochrane Database Syst Rev. 2012 Jul 11;7:CD004425. This update yielded six new trials but no change in conclusions. The six COCs evaluated in placebo-controlled trials are effective in reducing inflammatory and non-inflammatory facial acne lesions. Few important and consistent differences were found between COC types in their effectiveness for treating acne. How COCs compare to alternative acne treatments is unknown since only one trial addressed this issue. The use of standardized methods for assessing acne severity would help in synthesizing results across trials as well as aid in interpretation.
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Barbieri JS. **Isotretinoin** for Treatment of Acne. JAMA Dermatol. 2023 Oct 11. doi: 10.1001/jamadermatol.2023.2065.

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Health Canada Dec/15 is advising Canadians that the use of over-the-counter acne products applied to the skin containing **benzoyl peroxide** or **salicylic acid** may cause rare but **serious allergic reactions**.

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designed to create a verifiable link between the negative pregnancy test and the dispensing of the isotretinoin prescription to the female patient of childbearing potential. The iPLEDGE program requires that all patients meet qualification criteria and monthly program requirements. Before the patient receives his/her isotretinoin prescription each month, the prescriber must counsel the patient and document in the iPLEDGE system that the patient has been counseled about the risks of isotretinoin. There are also additional qualification criteria and monthly requirements for female patients of childbearing potential. As part of the ongoing risk management of isotretinoin products, it is crucial that a female of childbearing potential selects and commits to use two forms of effective contraception simultaneously for one month before, during, and for one month after isotretinoin therapy. She must have 2 negative urine or blood (serum) pregnancy tests with a sensitivity of at least 25 mIU/ml before receiving the initial isotretinoin prescription. The first pregnancy test is a screening test and can be conducted in the prescriber's office. The second pregnancy test must be done in a CLIA-certified laboratory according to the package insert. Each month of therapy, the patient must have a negative result from a urine or blood (serum) pregnancy test conducted by a CLIA-certified laboratory prior to receiving each prescription. https://www.ipledgeprogram.com/

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Nov 8, 2006 -- QLT Inc. announced positive results of a Phase IV clinical trial of **Aczone**(TM) dapsone in more than 50 patients with G6PD deficiency that was performed to meet a post-approval commitment requested by the FDA. Mar/08 FDA removes G6PD screening & labeling requirements from the label. June 6/08 /CNW/ - QLT Inc. (NASDAQ: QLTI; TSX: QLT) announced today that Health Canada has completed its review of QLT USA, Inc.'s labeling supplement (SNDS) for Aczone(R) and has **removed the glucose-6-phosphate dehydrogenase (G6PD) screening** and blood monitoring requirements.

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